

Safety data sheet

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BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 31.07.2019

Version: 1.0

Product: **GARDOBOND 24 TA**

(ID no. 30690901/SDS_GEN_00/EN)

Date of print 04.06.2021

1. Identification

Product identifier

GARDOBOND 24 TA

Recommended use: Phosphating solution for metal surfaces., Restricted to professional users.

Not recommended use: None known

Details of the supplier of the safety data sheet

Company:

Chemetall do Brasil Ltd

Rua Luiz Benezato, 100 - Jardim Ermida II

13212-161 - Jundiaí - SP, BRASIL

+ 55 11 4525-5324

sds.sa-chemetall@basf.com

Emergency telephone number

International emergency number:

Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (oral)

Resp. Sens. 1

Skin Sens. 1A

Muta. 2

Carc. 1A (by inhalation)

Repr. 1B (unborn child)

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STOT RE 1
Aquatic Acute 2
Aquatic Chronic 2
Met. Corr. 1
Skin Corr./Irrit. 1
Eye Dam./Irrit. 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System, UN (GHS)

Pictogram:



Signal Word:
Danger

Hazard Statement:

H314	Causes severe skin burns and eye damage.
H302	Harmful if swallowed.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317	May cause an allergic skin reaction.
H411	Toxic to aquatic life with long lasting effects.
H290	May be corrosive to metals.
H360	May damage the unborn child.
H372	Causes damage to organs through prolonged or repeated exposure.
H341	Suspected of causing genetic defects.
H350	May cause cancer by inhalation.

Precautionary Statements (Prevention):

P201	Obtain special instructions before use.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P264	Wash with plenty of water and soap thoroughly after handling.
P234	Keep only in original packaging.
P202	Do not handle until all safety precautions have been read and understood.
P270	Do not eat, drink or smoke when using this product.
P260	Do not breathe dust or mist.
P272	Contaminated work clothing should not be allowed out of the workplace.
P273	Avoid release to the environment.
P261	Avoid breathing dust/fume/gas/mist/vapours/spray.
P284	In case of inadequate ventilation wear respiratory protection.

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Precautionary Statements (Response):

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P304 + P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P302 + P352 IF ON SKIN: Wash with plenty of soap and water.

P363 Wash contaminated clothing before reuse.

P362 + P364 Take off contaminated clothing and wash it before reuse.

P330 Rinse mouth.

P310 Immediately call a POISON CENTER or doctor/physician.

P301 + P330 + P331 IF SWALLOWED: rinse mouth. Do NOT induce vomiting.

P333 + P313 If skin irritation or rash occurs: Get medical advice/attention.

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

P308 + P313 IF exposed or concerned: Get medical advice/attention.

P342 + P311 If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.

P391 Collect spillage.

P390 Absorb spillage to prevent material damage.

Precautionary Statements (Storage):

P405 Store locked up.

P406 Store in corrosive resistant container with a resistant inner liner.

Precautionary Statements (Disposal):

P501 Dispose of contents/container to hazardous or special waste collection point.

Other hazardsAccording to UN GHS criteria

Classified as 'corrosive' due to pH value. (Regulation (EC) No. 1272/2008)

3. Composition/Information on Ingredients**Substances**

Not applicable

MixturesChemical nature

inorganic
aqueous solution

Hazardous ingredients (GHS)

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According to UN GHS criteria

Nickel dinitrate

Content (W/W): $\geq 5\%$ - $< 7\%$

CAS Number: 13138-45-9

EC-Number: 236-068-5

Ox. Sol. 2

Acute Tox. 4 (Inhalation - dust)

Acute Tox. 4 (oral)

Skin Corr./Irrit. 2

Eye Dam./Irrit. 1

Resp. Sens. 1

Skin Sens. 1

STOT RE 1

Muta. 2

Carc. 1A (by inhalation)

Repr. 1B (unborn child)

Aquatic Acute 1

Aquatic Chronic 1

M-factor acute: 1

M-factor chronic: 1

H272, H318, H315, H334, H317, H360, H372,

H341, H350, H302 + H332, H400, H410

Specific concentration limit:Skin Corr./Irrit. 2: $\geq 20\%$ Skin Sens. 1: $\geq 0,01\%$ STOT RE 2: 0,1 - $< 1\%$ STOT RE 1: $\geq 1\%$

Manganese bis(dihydrogen phosphate)

Content (W/W): $\geq 7\%$ - $< 10\%$

CAS Number: 18718-07-5

EC-Number: 242-520-2

REACH registration number: 01-2119968560-32-0000

STOT RE 2 (Inhalation - dust)

Aquatic Chronic 3

Eye Dam./Irrit. 2A

H319, H412, H373

Orthophosphoric acid

Content (W/W): $\geq 7\%$ - $< 10\%$

CAS Number: 7664-38-2

EC-Number: 231-633-2

REACH registration number: 01-2119485924-24

Met. Corr. 1

Acute Tox. 4 (oral)

Eye Dam./Irrit. 1

Skin Corr./Irrit. 1B

H314, H302, H290

Specific concentration limit:Eye Dam./Irrit. 2: 10 - $< 25\%$ Skin Corr./Irrit. 1B: $\geq 25\%$ Skin Corr./Irrit. 2: 10 - $< 25\%$

Zinc bis (dihydrogen phosphate)

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Content (W/W): $\geq 10\%$ - $< 12,5\%$	Acute Tox. 4 (oral)
CAS Number: 13598-37-3	Aquatic Acute 1
EC-Number: 237-067-2	Aquatic Chronic 2
REACH registration number: 01-2119485974-19-0000	H302, H411, H400

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Immediately remove contaminated clothing. Remove affected person from danger area. First aid personnel should pay attention to their own safety. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

If inhaled:

Remove the affected individual into fresh air and keep the person calm. If symptoms persist, seek medical advice.

On skin contact:

Flush with copious amounts of water for at least 15 minutes. Immediate medical attention required.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist. Immediate medical attention required.

On ingestion:

Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting. Summon medical aid without delay.

Most important symptoms and effects, both acute and delayed

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11.

Hazards: May cause severe burns of the mouth and throat if orally ingested, as well as a danger of perforation of the oesophagus and the stomach.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:

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carbon dioxide, dry powder, alcohol-resistant foam, water spray

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

nitrogen oxides, phosphorus oxides

Advice for fire-fighters

Special protective equipment:

Appropriate breathing apparatus may be required.

Further information:

Cool closed containers in the vicinity of the source of fire. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures**Personal precautions, protective equipment and emergency procedures**

Use personal protective clothing. Advice on product handling can be found in sections 7 and 8 of this safety data sheet.

Environmental precautions

Do not allow to enter drains or waterways. Do not discharge into the subsoil/soil. If the product enters drains or sewers, the local water company should be contacted immediately; in the case of contamination of streams, rivers or lakes, the Environment Agency.

Methods and material for containment and cleaning up

Ensure adequate ventilation. Pick up with inert absorbent material (e.g. sand, earth etc.). Take up mechanically and collect in suitable container (adequately labelled) for disposal.

7. Handling and Storage**Precautions for safe handling**

Do not return residues to the storage containers. Provide good ventilation of working area (local exhaust ventilation if necessary). Avoid inhalation of vapour and spray mist. The workplace should be equipped with an emergency shower and eye-rinsing facility. Avoid contact with skin and eyes. Keep away from food, drink and animal feeding stuffs. Warn users about safety measures and precautions to prevent accidents.

Protection against fire and explosion:

The substance/product is non-combustible. The relevant fire protection measures should be noted.

Conditions for safe storage, including any incompatibilities

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Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Polyethyleneterephthalate (PET), Polypropylene (PP), Carbon steel (Iron), tinned carbon steel (Tinplate)

Further information on storage conditions: The entrance to storage rooms is to be granted only to appropriately trained personnel. Keep only in the original container. Keep in a cool, well-ventilated place. Avoid direct sunlight. avoid contact with metals Keep away from heat. Protect from frost.

Storage stability:

Storage temperature: 14 - 40 °C

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

18718-07-5: Manganese bis(dihydrogen phosphate)

7664-38-2: Orthophosphoric acid

Exposure controls

Personal protective equipment

Respiratory protection:

Wear respiratory protection if ventilation is inadequate. (Gas filter EN 14387 B)

Hand protection:

Chemical resistant protective gloves (EN 374)

chloroprene rubber (Neoprene)

Further information on penetration time is available from the manufacturer of the glove.

The gloves should be replaced immediately in case of damage or signs of wear. It is recommended to use preventative skin protection (skin cream).

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

Chemical resistant protective clothing according to DIN EN 13034 (Type 6)

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Eye wash fountains and safety showers must be easily accessible. Avoid contact with the skin, eyes and clothing. Take off immediately all contaminated clothing. Keep away from food, drink and animal feeding stuffs. Hands and/or face should be washed before breaks and at the end of the shift.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid
Colour:	green
Odour:	odourless
pH value:	< 2,0 (20 °C) (undiluted)
solidification temperature:	< 14 °C
onset of boiling:	not determined
Flash point:	not applicable
Flammability:	not applicable
Lower explosion limit:	not determined
Ignition temperature:	not determined
Vapour pressure:	(20 °C) not determined
Density:	(50 °C) not determined 1,410 g/cm ³ (20 °C)
Solubility in water:	fully soluble
Self ignition:	not self-igniting
Viscosity, kinematic:	(40 °C) not determined
Explosion hazard:	not explosive
Fire promoting properties:	not fire-propagating

Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Miscibility with water: miscible

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

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Corrosion to metals: Corrosive effect on metals.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

In aqueous solution, evolves hydrogen on contact with metals.

Conditions to avoid

Protect from sunlight. Avoid freezing. Avoid heat.

Incompatible materials

Substances to avoid:
reducing agents, metal

Hazardous decomposition products

Possible decomposition products:

When exposed to high temperatures hazardous decomposition products such as carbon monoxide, carbon dioxide, smoke, oxides of nitrogen may be produced., No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information**Information on toxicological effects**Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after single ingestion.

Irritation

Assessment of irritating effects:

Corrosive! Damages skin and eyes. May cause severe damage to the eyes.

Respiratory/Skin sensitization

Assessment of sensitization:

The substance may cause sensitization of the respiratory tract. Sensitization after skin contact possible.

Germ cell mutagenicity

Assessment of mutagenicity:

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Mutagenic properties can not be excluded on the basis of experimental data.

Carcinogenicity

Assessment of carcinogenicity:

The substance caused cancer by inhalation in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:

Based on available Data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:

The substance caused malformations/developmental toxicity in laboratory animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on available Data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated exposure to small quantities may affect certain organs.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

Toxic to aquatic life. Toxic to aquatic life with long lasting effects. There are no test results available for this product. Do not allow to enter drains or waterways.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

No data available concerning biodegradation and elimination.

Bioaccumulative potential

Bioaccumulation potential:

No data available.

Mobility in soil

Assessment transport between environmental compartments:

Adsorption in soil: No data available.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

Contaminated packaging:

Containers which are not properly emptied must be disposed pursuant to Directive 2008/98/EC

Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Land transport

ADR

UN number	UN3264
UN proper shipping name:	CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains Orthophosphoric acid)
Transport hazard class(es):	8, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	Tunnel code: E

RID

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UN number: UN3264
UN proper shipping name: CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains Orthophosphoric acid)
Transport hazard class(es): 8, EHSM
Packing group: III
Environmental hazards: yes
Special precautions for user:

Inland waterway transport

ADN

UN number: UN3264
UN proper shipping name: CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains Orthophosphoric acid)
Transport hazard class(es): 8, EHSM
Packing group: III
Environmental hazards: yes
Special precautions for user:

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number: UN 3264
UN proper shipping name: CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains Orthophosphoric acid, Zinc bis (dihydrogen phosphate))
Transport hazard class(es): 8, EHSM
Packing group: III
Environmental hazards: yes
Marine pollutant: YES
Special precautions for user:

Air transport

IATA/ICAO

UN number: UN 3264
UN proper shipping name: CORROSIVE LIQUID, ACIDIC, INORGANIC, N.O.S. (contains Orthophosphoric acid)
Transport hazard class(es): 8

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Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	

Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Acute Tox.	Acute toxicity
Resp. Sens.	Respiratory sensitization
Skin Sens.	Skin sensitization
Muta.	Germ cell mutagenicity
Carc.	Carcinogenicity
Repr.	Reproductive toxicity
STOT RE	Specific target organ toxicity — repeated exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Met. Corr.	Corrosive to metals
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Ox. Sol.	Oxidising solids
H272	May intensify fire; oxidizer.
H318	Causes serious eye damage.
H315	Causes skin irritation.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317	May cause an allergic skin reaction.
H360	May damage the unborn child.
H372	Causes damage to organs through prolonged or repeated exposure.
H341	Suspected of causing genetic defects.
H350	May cause cancer by inhalation.
H302 + H332	Harmful if swallowed or if inhaled
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H319	Causes serious eye irritation.

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H412	Harmful to aquatic life with long lasting effects.
H373	May cause damage to organs through prolonged or repeated exposure.
H314	Causes severe skin burns and eye damage.
H302	Harmful if swallowed.
H290	May be corrosive to metals.
H411	Toxic to aquatic life with long lasting effects.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.